

## Heat Capacity 1

Essential Pre-Uni Physics G3.1

Subject & topics: Physics | Thermal | Heat      Stage & difficulty: GCSE P1, A Level P1

Data:

- Specific heat capacity of water:  $4180 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of aluminium:  $880 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of iron:  $435 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of paraffin oil:  $2130 \text{ J kg}^{-1} \text{ K}^{-1}$

Complete the values in the table below. Give your answers to 2 sf.

| Energy / J           | Material  | Mass / kg | Initial temperature / °C | Final temperature / °C |
|----------------------|-----------|-----------|--------------------------|------------------------|
| <input type="text"/> | Aluminium | 0.290     | 15                       | 82                     |
| 45200                | Paraffin  | 2.30      | 3.0                      | <input type="text"/>   |
| 81000                | Water     | 1.50      | 11                       | <input type="text"/>   |

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STEM SMART Physics 27 - Heat and Mixtures

## Heat Capacity 3

### Essential Pre-Uni Physics G3.3

**Subject & topics:** Physics | Thermal | Heat

**Stage & difficulty:** GCSE C2, A Level P1

- Specific heat capacity of water:  $4180 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of aluminium:  $880 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of iron:  $435 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of paraffin oil:  $2130 \text{ J kg}^{-1} \text{ K}^{-1}$

These specific heat capacities can also be found within the hint tabs.

How much water can a shower head heat each second from  $12^\circ\text{C}$  to  $41^\circ\text{C}$  if the heater has a power of  $4200 \text{ W}$ ? Assume that no heat is lost to the surroundings, and give your answer in kilograms.

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## Heat Capacity 4

### Essential Pre-Uni Physics G3.4

**Subject & topics:** Physics | Thermal | Heat

**Stage & difficulty:** GCSE C2, A Level P1

- Specific heat capacity of water:  $4180 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of aluminium:  $880 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of iron:  $435 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of paraffin oil:  $2130 \text{ J kg}^{-1} \text{ K}^{-1}$

These specific heat capacities can also be found within the hint tabs.

If  $0.024 \text{ kg}$  of water gets trapped in the shower heater (the heater has a power of  $4200 \text{ W}$ ) of [question G3.3](#), the thermal sensor must stop the current before the water reaches  $80^\circ\text{C}$ . Assuming that the water is at  $35^\circ\text{C}$  when the fault occurs, how quickly must the thermal sensor act? Give your answer in seconds.

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## Heat Capacity 8

### Essential Pre-Uni Physics G3.8

**Subject & topics:** Physics | Thermal | Heat

**Stage & difficulty:** GCSE C3, A Level P2

- Specific heat capacity of water:  $4180 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of aluminium:  $880 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of iron:  $435 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of paraffin oil:  $2130 \text{ J kg}^{-1} \text{ K}^{-1}$

These specific heat capacities can also be found within the hint tabs.

If I add 210 g of rivets at  $303^\circ\text{C}$  made of some unknown metal to 500 g of water at  $15^\circ\text{C}$ , and the final temperature is  $34^\circ\text{C}$ , what is the specific heat capacity of the mystery metal? Give your answer to 2 significant figures.

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## Latent Heat and Heat Capacity 1

### Essential Pre-Uni Physics G4.1

**Subject & topics:** Physics | Thermal | Heat**Stage & difficulty:** GCSE C1, A Level P1

- Specific heat capacity of water:  $4180 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of ice:  $2030 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific latent heat of fusion of ice:  $3.35 \times 10^5 \text{ J kg}^{-1}$
- Specific latent heat of vaporization of water:  $2.26 \times 10^6 \text{ J kg}^{-1}$

In all questions, assume that the heat capacities given above remain constant at all temperatures.

#### Part A

##### Frozen pipe

A frozen pipe contains 5.60 kg of ice. How much energy is needed to melt it without changing its temperature?

#### Part B

##### Warming and melting

If the ice were initially at  $-3.5^\circ\text{C}$ , how much energy would be taken to warm it to melting point and then melt it?

Question deck:

**STEM SMART Physics 27 - Heat and Mixtures**

## Latent Heat and Heat Capacity 2

### Essential Pre-Uni Physics G4.2

**Subject & topics:** Physics | Thermal | Heat**Stage & difficulty:** GCSE C2, A Level P1

- Specific heat capacity of water:  $4180 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of ice:  $2030 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific latent heat of fusion of ice:  $3.35 \times 10^5 \text{ J kg}^{-1}$
- Specific latent heat of vaporization of water:  $2.26 \times 10^6 \text{ J kg}^{-1}$

In all questions, assume that the heat capacities given above remain constant at all temperatures.

#### Part A

##### Initial temperature

A certain quantity of ice requires  $10.0 \text{ J}$  to warm it to melting temperature. It then requires  $100 \text{ J}$  to melt it. Calculate the initial temperature of the ice in  $^{\circ}\text{C}$ , assuming no heat loss to the surroundings.

#### Part B

##### Final temperature

The water at freezing point in Part A is then heated using a further  $100 \text{ J}$ . What is its final temperature? Give your answer in  $^{\circ}\text{C}$

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## Latent Heat and Heat Capacity 5

Essential Pre-Uni Physics G4.5

**Subject & topics:** Physics | Thermal | Heat

**Stage & difficulty:** GCSE C3, A Level C2

- Specific heat capacity of water:  $4180 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific heat capacity of ice:  $2030 \text{ J kg}^{-1} \text{ K}^{-1}$
- Specific latent heat of fusion of ice:  $3.35 \times 10^5 \text{ J kg}^{-1}$
- Specific latent heat of vaporization of water:  $2.26 \times 10^6 \text{ J kg}^{-1}$

In all questions, assume that the heat capacities given above remain constant at all temperatures.

A mass of  $0.35 \text{ kg}$  of ice at  $-15^\circ\text{C}$  is lowered into an insulated beaker containing  $0.61 \text{ kg}$  of water at  $59^\circ\text{C}$ .

Part A

### Equilibrium temperature

What is the temperature after equilibrium has been reached? Give your answer in  $^\circ\text{C}$

Part B

### Minimum mass of water for $0.0^\circ\text{C}$

What is the minimum mass of water at  $59^\circ\text{C}$  needed in the beaker to achieve a final temperature of  $0.0^\circ\text{C}$ ?

Part C

**Maximum mass of water for  $0.0^{\circ}\text{C}$**

What is the maximum mass of water at  $59^{\circ}\text{C}$  that could be present in the beaker to achieve a final temperature of  $0.0^{\circ}\text{C}$ ?

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## Sea Level Rise

**Subject & topics:** Physics | Thermal | Heat**Stage & difficulty:** A Level C2

### Part A

#### Ocean heating

Sea level is currently observed to increase at a total rate of about  $3 \text{ mm/year}$ . Out of this total rate, approximately  $1 \text{ mm/year}$  is due to thermal expansion of the warming sea water. This is known as steric sea level change.

Assume that the ocean heating occurs uniformly over the top  $1000 \text{ m}$  of the ocean at a rate of  $0.01 \text{ }^\circ\text{C year}^{-1}$ . Calculate the power required for this ocean heating.

Use the following information:

- Assume that the Earth is a perfect sphere with radius  $6371 \text{ km}$
- The oceans cover  $70 \%$  of the Earth's surface
- The density of sea water is  $1025 \text{ kg m}^{-3}$
- The heat capacity of sea water  $C_p = 4.006 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

### Part B

#### Melting ice on land

For this question, we assume that the remaining  $2 \text{ mm year}^{-1}$  of sea level change occurs due to the melting of land-based ice.

Estimate the rate of melting of land-based ice (in  $\text{kg year}^{-1}$ ) needed to achieve the observed rate of sea level increase due to the **non-steric** effect. Assume that the area of the ocean remains constant.

### Part C

#### Power of melting

Estimate the power required to account for the observed rate of melting, assuming that the ice is initially at  $T_{\text{ice}} = -20^\circ\text{C}$ .

The heat capacity of ice is  $C_{\text{ice}} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$  and the latent heat of fusion of ice is  $L_{\text{fusion}} = 330 \text{ kJ kg}^{-1}$ .

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